An NTU-Based Numerical Assessment of CO2 Loading Times in a Recirculating Hollow-Fiber Membrane Contactor

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Abstract

This numerical study assesses CO_2 loading-time predictions for a recirculating alkaline hollow-fiber membrane contactor considered for direct air capture. Perfect-sink and equilibrium-aware number-of-transfer-units (NTU) models are compared with a finite-pass model using the same prescribed mass-transfer coefficient and constant volumetric gas flow. A dimensionless chemical burden factor summarizes the equilibrium contribution to loading time. At the baseline pH 10.00 target, the finite-pass model predicts 19.35% more loading time than the perfect-sink model and 0.78% more than the equilibrium-aware model. At a normalized remaining driving-force fraction of 0.05, the additional time relative to the equilibrium-aware prediction increases from 1.13% at the baseline liquid flow to 62.67% at 0.03 times that flow. Numerical refinement and analytical checks support the consistency of these calculations. Within-pass loading increasingly affects predictions as liquid flow decreases and the target approaches equilibrium.

1. Objective and Scope

For a recirculating alkaline hollow-fiber membrane contactor (HFMC) considered for direct air capture (DAC), a useful performance measure is the time required to reach a specified liquid loading state. A number-of-transfer-units (NTU) description provides a compact representation of gas depletion through the contactor. The predicted loading time also depends on how the model represents the evolving liquid composition and the associated reduction in CO_2 absorption driving force.

Previous cross-flow HFMC studies have considered concentration changes in both phases and, for reactive absorption, reversible chemical reactions and equilibria [1,2]. This report asks how closely an equilibrium-aware NTU model that assigns the current tank composition throughout the contactor reproduces finite-pass loading-time predictions, and how the remaining differences depend on liquid flow and target proximity to equilibrium.

Three matched formulations are compared: perfect-sink (PS) and equilibrium-aware (EA) NTU models and a finite-pass (FP) liquid-loading model. The equilibrium contribution is summarized by the dimensionless chemical burden factor B_{chem}, calculated from the loading-time integral. The main comparisons vary liquid flow and target proximity while retaining the same prescribed mass-transfer

coefficient, carbonate equilibrium relations, geometry, liquid inventory, and volumetric gas flow. The scope is to separate tank-equilibrium effects from within-pass liquid loading and assess the uniform-liquid approximation under these assumptions.

2. Model Formulation

2.1. System Configuration and Common Assumptions

The system comprises a well-mixed alkaline liquid reservoir recirculated through a hollow-fiber membrane contactor (HFMC), with a separate CO_2-containing gas feed. The reference fiber bundle comprises 20 rows of 20 fibers, with all 400 fiber lumens connected in parallel on the liquid side. Gas passes sequentially across the rows, while each liquid path receives the common tank composition and returns to the reservoir. The present model represents the discrete rows as a continuous distribution of liquid paths along the gas-flow direction. Membrane area and liquid flow are distributed in the same proportion along this coordinate. The total membrane area is calculated from the outer fiber surface as $A = \pi d_o L N_f$. Table 1 lists the numerical baseline adopted from the model configuration. The nominal outer diameter is calculated from the Q3/2 inner diameter (600 μm) and wall thickness (200 μm) reported by 3M [3]. Full-precision inputs are retained in Appendix D. The system configuration and model coordinates are illustrated in Figure 1.

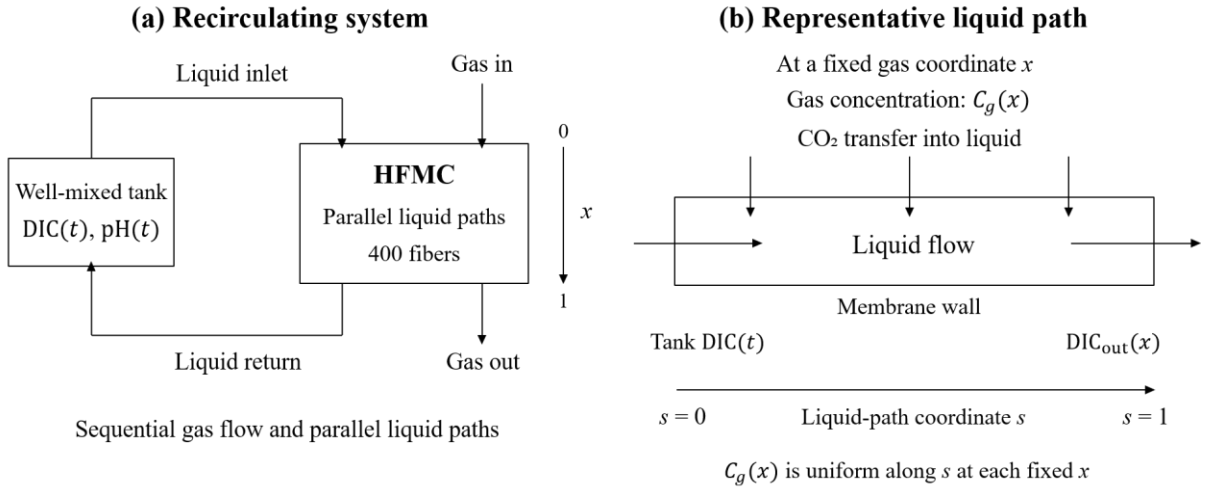


Figure 1. Recirculating HFMC System and Model Coordinates. (a) A well-mixed tank supplies 400 liquid-parallel fibers. The discrete fiber rows are represented continuously along the gas coordinate x. (b) A representative liquid path at fixed x, with normalized membrane-area coordinate s. Each path receives the tank DIC, and $C_g(x)$ is assumed uniform along s at fixed x.

All three models assume constant temperature, pressure, and liquid inventory. Carbonate speciation is described by an ideal-concentration equilibrium model at fixed alkalinity; activity corrections and explicit reaction kinetics are not included. Henry's law gives the local equilibrium dissolved CO_2 concentration as

$$C^*_{\mathrm{local}} = Hp_{\mathrm{CO_2,local}},$$

where H is the Henry solubility coefficient and $p_{\mathrm{CO_2,local}}$ is the local gas-phase CO_2 partial pressure. The Henry coefficient in Table 1 follows Sander [4]. The local CO_2 absorption flux is written as

$$J = K\left(C^*_{\mathrm{local}} - C_{\mathrm{CO_2},L}\right)$$

where J is positive for transfer from the gas into the liquid and is expressed per unit outer membrane area, and $C_{\mathrm{CO_2},L}$ is the local dissolved CO_2 concentration in the liquid. Because both concentrations in the driving force are expressed on a liquid-phase basis, K is an overall liquid-basis mass-transfer coefficient. Its value is prescribed and held fixed throughout the model comparisons. The gas is treated as ideal, and its volumetric flow is specified at the listed temperature and pressure. CO_2 depletion is retained, but the associated gas-flow contraction is neglected in the main comparisons, so all three models use the same constant volumetric gas flow.

The tank composition evolves in time, whereas the contactor is treated as quasi-steady at each tank state, without a separate transient storage balance within the module. The common tank balance is

$$V_L \frac{d\mathrm{DIC}}{dt} = \dot{n}_{\mathrm{abs}}(\mathrm{DIC}), \qquad \mathrm{DIC}(0) = \mathrm{DIC}_0,$$

where DIC is the tank dissolved inorganic carbon concentration, V_L is the well-mixed liquid volume, and $\dot{n}_{\mathrm{abs}}$ is the net CO_2 absorption rate into the liquid. The perfect-sink, equilibrium-aware, and finite-pass models differ in how this absorption rate is evaluated, as defined in Sections 2.3 and 2.4.

2.2. Chemical State, Equilibrium, and Target Definition

The liquid chemical state is described by the DIC concentration and alkalinity,

$$\mathrm{DIC} = [\mathrm{CO_2(aq)}] + [\mathrm{HCO_3^-}] + [\mathrm{CO_3^{2-}}],$$

$$\mathrm{Alk} = [\mathrm{HCO_3^-}] + 2[\mathrm{CO_3^{2-}}] + [\mathrm{OH^-}] - [\mathrm{H^+}],$$

where brackets denote molar concentrations. At the prescribed temperature and fixed alkalinity, the carbonate equilibrium relations determine pH and the dissolved CO_2 concentration from DIC [5]. The latter is denoted by

$$C_{\mathrm{CO_2},L} \equiv [\mathrm{CO_2(aq)}] = f(\mathrm{DIC};\mathrm{Alk}).$$

This dissolved CO_2 concentration determines the liquid contribution to the mass-transfer driving force. The same chemical equilibrium relations are used to calculate pH from DIC in all three models. Detailed speciation equations, equilibrium constants, and concentration conventions are provided in Appendix A.1.

Using the Henry-law convention in Section 2.1, the inlet-gas reference concentration and the corresponding equilibrium DIC are defined by

$$C_{\mathrm{in}}^{*} = HPy_{\mathrm{CO_2,in}}, \qquad f\left(\mathrm{DIC_{eq}}; \mathrm{Alk}\right) = C_{\mathrm{in}}^{*}.$$

Here and in the endpoint expressions below, C_{in}^{*} denotes the inlet-gas value; local Henry-law equilibrium concentrations within the contactor depend on the local gas composition.

A target state $\mathrm{DIC_{tar}}$ is specified either by pH or by the dimensionless remaining driving-force fraction,

$$\delta(\mathrm{DIC}) = \frac{C_{\mathrm{in}}^{*} - f(\mathrm{DIC}; \mathrm{Alk})}{C_{\mathrm{in}}^{*}}, \qquad \delta_{\mathrm{tar}} = \delta(\mathrm{DIC_{tar}}).$$

Smaller δ_{tar} indicates closer approach to equilibrium with the inlet gas, with $\delta_{\mathrm{tar}} = 0$ at $\mathrm{DIC_{tar}} = \mathrm{DIC_{eq}}$. A prescribed pH target is converted to $\mathrm{DIC_{tar}}$ through the same chemical equilibrium relations. Comparisons are restricted to $\mathrm{DIC_0} < \mathrm{DIC_{tar}} < \mathrm{DIC_{eq}}$, so that each target lies above the initial loading and below the equilibrium limit. The target sets are specified in Section 3.1. The subscript tar marks the target.

2.3. Perfect-Sink and Equilibrium-Aware NTU Models

For the ideal gas feed, the inlet CO_2 concentration and the dimensionless Henry partition coefficient are

$$C_{g,\mathrm{in}} = \frac{Py_{\mathrm{CO_2,in}}}{RT}, \qquad m = \frac{C_{\mathrm{in}}^{*}}{C_{g,\mathrm{in}}} = HRT,$$

Classical transfer-unit formulations retain the equilibrium driving force [6]. The perfect-sink approximation neglects dissolved CO_2 in the local driving force while retaining gas depletion. Integrating the constant-flow gas balance over the outer membrane area gives the transfer-unit group and module absorption rate:

$$\mathrm{NTU}_g = \frac{mKA}{Q_g}.$$

$$\dot{n}_{\mathrm{PS}} = Q_g\left(C_{g,\mathrm{in}} - C_{g,\mathrm{out}}\right) = Q_g C_{g,\mathrm{in}}\left(1 - e^{-\mathrm{NTU}_g}\right).$$

Because this absorption rate is constant under the stated assumptions, the time required to reach the target DIC is

$$t_{PS} = \frac{V_L(\mathrm{DIC_{tar}} - \mathrm{DIC_0})}{\dot{n}_{PS}}.$$

At each tank state, the equilibrium-aware model assigns the same dissolved CO_2 concentration throughout the contactor. Its gas balance then reduces the perfect-sink absorption rate by the remaining driving-force fraction:

$$\dot{n}_{\mathrm{EA}}(\mathrm{DIC}) = \dot{n}_{\mathrm{PS}}\left[1 - \frac{f(\mathrm{DIC};\mathrm{Alk})}{C^*_{\mathrm{in}}}\right] = \dot{n}_{\mathrm{PS}}\,\delta(\mathrm{DIC}).$$

Integrating this state-dependent rate in the tank balance gives the target time and the chemical burden factor:

$$t_{\mathrm{EA}} = t_{\mathrm{PS}} B_{\mathrm{chem}}, \qquad B_{\mathrm{chem}} = \frac{1}{\mathrm{DIC_{tar}} - \mathrm{DIC_0}} \int_{\mathrm{DIC_0}}^{\mathrm{DIC_{tar}}} \frac{d\mathrm{DIC}}{\delta(\mathrm{DIC})}.$$

The dimensionless chemical burden factor B_{chem} is therefore the average inverse normalized driving force over the DIC loading interval. It follows directly from the loading-time integral. For the reachable absorption states considered here, $B_{\mathrm{chem}} \geq 1$, approaching unity when $f(\mathrm{DIC};\mathrm{Alk}) \ll C^*_{\mathrm{in}}$ throughout the interval. Additional derivation details are provided in Appendix A.2.

Although t_{EA} accounts for the evolving tank composition, it neglects additional liquid loading within an individual contactor pass. Consequently, neither t_{PS} nor t_{EA} depends on Q_l when the geometry, gas conditions, liquid inventory, chemistry, target, and prescribed K are held fixed. The finite-pass model in Section 2.4 relaxes this uniform-liquid assumption.

2.4. Finite-Pass Liquid-Loading Model

At each fixed tank state, the finite-pass model allows DIC to increase along the parallel liquid paths while gas-phase CO_2 is depleted sequentially. Let $x \in [0,1]$ denote the normalized gas coordinate and $s \in [0,1]$ the normalized membrane-area coordinate along a liquid path. An interval dx is assigned membrane area $A\,dx$ and liquid flow $Q_l\,dx$, consistent with the allocation in Section 2.1. All liquid paths receive the current tank composition.

At a given x, the gas CO_2 concentration $C_g(x)$ is taken as uniform along s. The local absorption flux is

$$J(x,s) = K\big[mC_g(x) - f(\mathrm{DIC}_L(x,s);\mathrm{Alk})\big],$$

where DIC_L is the DIC concentration within the liquid path. The liquid-pass balance and inlet condition are

$$\frac{\partial \mathrm{DIC}_L(x,s)}{\partial s} = \frac{A}{Q_l} J(x,s), \qquad \mathrm{DIC}_L(x,0) = \mathrm{DIC}.$$

Defining $\mathrm{DIC}_{\mathrm{out}}(x) = \mathrm{DIC}_L(x,1)$, the gas balance under constant volumetric gas flow is

$$Q_g \frac{dC_g}{dx} = -Q_l[\mathrm{DIC}_{\mathrm{out}}(x) - \mathrm{DIC}], \qquad C_g(0) = C_{g,\mathrm{in}}.$$

The coupled balances give the module absorption rate,

$$\dot{n}_{FP}(\mathrm{DIC}) = Q_g\big[C_{g,\mathrm{in}} - C_g(1)\big] = Q_l \int_0^1 [\mathrm{DIC}_{\mathrm{out}}(x) - \mathrm{DIC}]\, dx.$$

All contactor concentrations in these expressions are evaluated at the prescribed tank DIC. Coupling this rate to the tank balance gives the target loading time,

$$t_{FP} = V_L \int_{\mathrm{DIC}_0}^{\mathrm{DIC}_{\mathrm{tar}}} \frac{d\mathrm{DIC}}{\dot{n}_{FP}(\mathrm{DIC})}.$$

Liquid flow enters through KA/Q_l in the liquid-pass balance. When the dissolved CO_2 concentration changes negligibly within each pass, the formulation approaches the equilibrium-aware model in Section 2.3. Numerical integration is described in Section 3.1, with detailed settings and refinement studies in Appendices B.1 and B.2.

3. Numerical Methods and Verification

3.1. Reference Conditions and Numerical Methods

Table 1. Baseline Inputs for the Matched Model Comparisons.

Parameter	Symbol	Baseline value	Unit
Number of fibers	N_f	400	—
Active fiber length	L	0.20	m
Fiber outer diameter	d_o	1.00	mm
Total outer membrane area	A	0.251327	m²
Well-mixed liquid volume	V_L	0.500	L
Liquid recirculation flow rate	Q_l	271.43	mL min⁻¹
Gas volumetric flow rate	Q_g	234	L min⁻¹
Temperature	T	298.15	K
Absolute pressure	P	101325	Pa
Inlet CO_2 mole fraction	$y_{CO_2,in}$	400	ppm
Initial alkalinity	Alk	100	mol m⁻³
Initial DIC concentration	DIC_0	0	mol m⁻³
Overall liquid-basis mass-transfer coefficient	K	1.0×10^{-3}	m s⁻¹
Henry solubility coefficient	H	3.4×10^{-4}	mol m⁻³ Pa⁻¹

The baseline comparisons use the inputs in Table 1 and four target pH values: 11.00, 10.10, 10.05, and 10.00. The liquid-flow sweep uses $Q_l/Q_{l,\mathrm{ref}} = 0.03, 0.1, 0.3, 1, 3$, and 10, where $Q_{l,\mathrm{ref}}$ is the baseline liquid flow rate. Each flow rate is evaluated at $\delta_{\mathrm{tar}} = 0.80, 0.50, 0.15, 0.05$, and 0.01, giving 30 case–target combinations. Other inputs are held fixed during this sweep, including K and Q_g. Selected additional variations in alkalinity, inlet CO_2, initial DIC, and gas flow extend the numerical-refinement set to 100 case–target combinations. The exact combinations are listed in Appendix B.2, with supplementary chemistry comparisons in Appendix C.1.

The liquid-pass and gas-coordinate balances are integrated numerically, and loading times are evaluated by quadrature after transforming the distance to equilibrium. Appendix B.1 specifies the solver controls, endpoint resolutions, and chemical-state calculation.

The pH trajectories are assembled from target-time calculations at successively lower pH values, with each time integrated from the common initial state. Plotting-grid resolution is checked separately from the numerical integration, as documented in Appendix B.2.

For the main model comparisons, the relative time difference is defined as

$$E_i = 100\left(\frac{t_{FP}}{t_i} - 1\right), \qquad i \in \{PS, EA\}.$$

The denominator is the corresponding simplified-model time. A positive E_i therefore expresses the additional loading time predicted by the finite-pass model relative to that simplified prediction.

3.2. Numerical Verification

Numerical verification combines joint resolution refinement over 100 case–target combinations, a further refinement of ten selected combinations, analytical rate benchmarks, independent liquid-pass checks, and limiting-model recovery. The exact case sets and methods are given in Appendix B.

Table 2 summarizes the maximum absolute percentage differences, calculated as $100|X_{\mathrm{test}}/X_{\mathrm{ref}} - 1|$. For refinement checks, X is t_{FP} and the reference is the finer calculation. For the remaining checks, the reference is the analytical expression or alternative implementation identified in the table. Detailed refinement results are provided in Appendix B.2, and benchmark definitions and chemical-equilibrium consistency checks are provided in Appendix B.3.

Table 2. Summary of Numerical Refinement and Consistency Checks.

Check	Compared quantity and scope	Maximum difference (%)
Joint coarse-to-fine refinement	t_{FP}; 100 case–target combinations	1.31×10^{-3}
Additional fine-to-finer refinement	t_{FP}; 10 selected case–target combinations	7.22×10^{-5}
Linear-equilibrium analytical benchmark	Module absorption rate versus the exact solution for a linear liquid CO_2–DIC relation; 3 tank states	8.87×10^{-9}
Liquid-pass implementation cross-check	Single-pass DIC increase from Runge–Kutta integration versus an integral-balance solver; 9 conditions	1.03×10^{-6}
Uniform-liquid recovery	Target time from the uniform-liquid contactor calculation versus t_{EA}; 9 case–target combinations	4.22×10^{-13}

4. Results and Interpretation

4.1. Baseline Comparison of Target Loading Times

Figure 2 compares the target loading times predicted by the perfect-sink NTU model (t_{PS}), the equilibrium-aware NTU model (t_{EA}), and the finite-pass model (t_{FP}) at the baseline conditions. All three models use the same prescribed mass-transfer coefficient and constant volumetric gas flow. The four target pH values, 11.00, 10.10, 10.05, and 10.00, correspond to progressively smaller remaining driving-force fractions, from $\delta_{\mathrm{tar}} = 0.984$ to 0.159. At pH 11.00, the predicted times are nearly identical: 2.236, 2.237, and 2.238 h, respectively. At pH 10.00, the predictions separate to 3.120, 3.695, and 3.724 h (Figure 2a,b).

To distinguish the two approximations, Figure 2c reports $E_i = 100(t_{FP}/t_i - 1)$ for $i = PS, EA$, using each simplified-model time as the denominator. The additional time relative to the perfect-sink prediction increases from 0.071% at pH 11.00 to 19.35% at pH 10.00. Including the liquid equilibrium driving force substantially reduces this difference: E_{EA} ranges from 0.012% to 0.780% across the four targets. At pH 10.00, the remaining finite-pass contribution is approximately 104 s, compared with an equilibrium-aware loading time of 3.695 h.

At the baseline liquid flow, tank equilibrium explains most of the difference from the perfect-sink prediction. The remaining finite-pass contribution is below 1% of the equilibrium-aware time for these four targets.

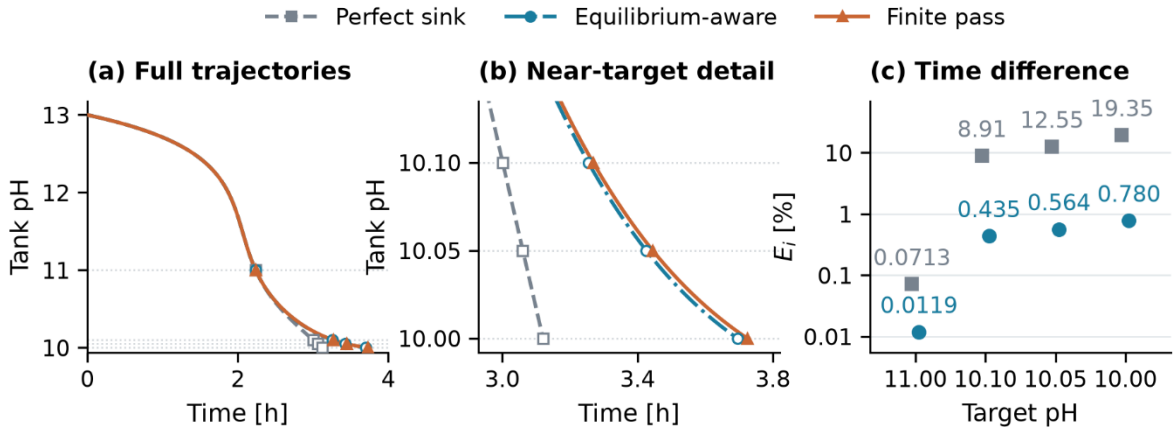


Figure 2. Baseline Loading-Time Predictions. (a) Tank pH trajectories for the perfect-sink, equilibrium-aware, and finite-pass models. (b) Expanded view of the three lowest target pH values. Markers indicate target crossings. (c) Relative time differences, $E_i = 100(t_{\mathrm{FP}}/t_i - 1)$, for $i = PS, EA$, on a logarithmic vertical axis. Baseline conditions are listed in Table 1, with constant gas volumetric flow and fixed K.

4.2. Effects of Liquid Flow and Equilibrium Proximity

Figure 3 examines whether the baseline agreement persists when the liquid flow rate and target proximity to equilibrium are varied. The liquid flow rate was varied from 0.03 to 10 times the baseline value of 271.43 mL min^{-1}, with five target driving-force fractions, $\delta_{\mathrm{tar}} = 0.80$, 0.50, 0.15, 0.05, and 0.01. Chemistry, geometry, liquid inventory, the prescribed mass-transfer coefficient, and volumetric gas flow were held fixed. Smaller δ_{tar} denotes a smaller remaining driving force.

At $\delta_{\mathrm{tar}} = 0.05$, t_{PS} and t_{EA} remain at 3.163 and 4.145 h, respectively, throughout the flow sweep (Figure 3a). In contrast, t_{FP} increases from 4.192 h at the baseline flow to 4.700 h at one-tenth of that flow and 6.742 h at 0.03 times the baseline flow. The corresponding relative time differences, $E_{\mathrm{EA}} = 100(t_{FP}/t_{\mathrm{EA}} - 1)$, are 1.13%, 13.38%, and 62.67%. At ten times the baseline flow, this difference falls to 0.11%.

The residual difference also increases as the target approaches equilibrium (Figure 3b). At one-tenth of the baseline flow, E_{EA} rises from 1.83% at $\delta_{\mathrm{tar}} = 0.80$ to 17.75% at $\delta_{\mathrm{tar}} = 0.01$; at the baseline flow, the corresponding values are 0.13% and 1.52%. The finite-pass contribution increases at lower liquid flow and closer-to-equilibrium targets.

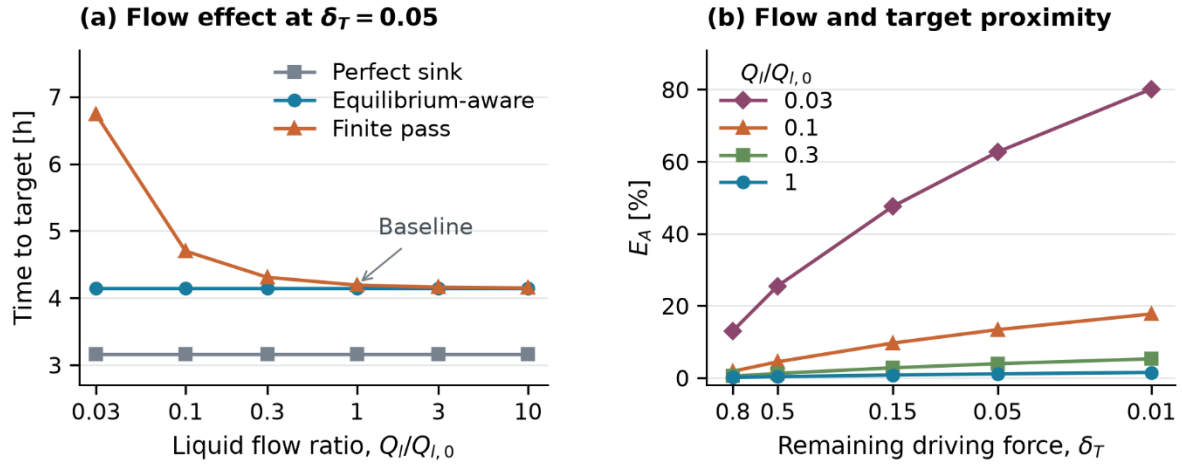


Figure 3. Effects of Liquid Flow and Target Proximity on Loading-Time Predictions. (a) Target times at $\delta_{\mathrm{tar}} = 0.05$ versus $Q_l/Q_{l,\mathrm{ref}}$. (b) Relative time difference, $E_{\mathrm{EA}} = 100(t_{\mathrm{FP}}/t_{\mathrm{EA}} - 1)$, versus δ_{tar} for selected liquid flows. Markers indicate calculated cases. Horizontal axes are logarithmic; smaller δ_{tar} indicates closer approach to equilibrium. Other inputs follow Table 1, with fixed K and gas flow.

4.3. Physical Interpretation and Model Applicability

The comparisons in Figures 2 and 3 distinguish two contributions to the departure from the perfect-sink prediction. The difference between t_{PS} and t_{EA} reflects the reduction in driving force as the dissolved molecular CO_2 concentration increases during tank loading. The additional difference between t_{EA} and t_{FP} reflects liquid loading within individual contactor passes. Although the equilibrium-aware model

updates the tank chemical state throughout loading, it assigns that bulk state to the liquid throughout the contactor at each instant. The finite-pass model instead allows DIC to increase along each liquid flow path, reducing the local driving force further.

The baseline agreement should therefore be assessed at the intended liquid flow and target against a stated acceptable loading-time error. Appendix C.3 relates these trends to a first-order small-pass correction.

5. Conclusions and Limitations

The equilibrium-aware NTU model captures most of the baseline increase in loading time over the perfect-sink estimate. At the pH 10.00 target, the finite-pass time is 19.35% above the perfect-sink time but only 0.78% above the equilibrium-aware time. The chemical burden factor represents the tank-equilibrium contribution.

The finite-pass contribution grows as liquid flow falls and the target approaches equilibrium: at a remaining driving-force fraction of 0.05, it reaches 62.67% relative to the equilibrium-aware time at 0.03 times the baseline liquid flow. These predictions use ideal equilibrium chemistry, a prescribed flow-independent transfer coefficient, and a quasi-steady constant-volumetric-flow contactor. Numerical checks do not replace experimental validation.

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Appendix A. Chemical Equilibrium and NTU Derivation

A.1. Carbonate Equilibrium and Target Conversion

The chemical closure uses ideal concentration-based equilibria. All concentrations are expressed in mol m⁻³. Defining $h = [\mathrm{H}^+]$ and the reference concentration $c^\circ = 10^3\ \mathrm{mol\,m^{-3}}$, the model pH is

$$\mathrm{pH} = -\log_{10}(h/c^\circ), \qquad h = c^\circ 10^{-\mathrm{pH}}.$$

The equilibrium constants are defined by

$$K_1 = \frac{h[\mathrm{HCO_3^-}]}{[\mathrm{CO_2(aq)}]}, \qquad K_2 = \frac{h[\mathrm{CO_3^{2-}}]}{[\mathrm{HCO_3^-}]}, \qquad K_w = h[\mathrm{OH^-}].$$

The prescribed values used in the calculations at 298.15 K are $K_1 = 4.45 \times 10^{-4}\ \mathrm{mol\,m^{-3}}$ [7], $K_2 = 4.69 \times 10^{-8}\ \mathrm{mol\,m^{-3}}$ [8], and $K_w = 1.00 \times 10^{-8}\ \mathrm{mol^2\,m^{-6}}$ [9]. These literature-based numerical values are adopted as ideal concentration-form constants on the stated unit basis. The dissolved CO_2 component is represented by $\mathrm{CO_2(aq)}$; no separate carbonic-acid state or hydration kinetics is included.

The fractions of DIC present as dissolved CO_2, bicarbonate, and carbonate are, respectively,

$$\alpha_0(h) = \frac{h^2}{h^2 + K_1 h + K_1 K_2}, \qquad \alpha_1(h) = \frac{K_1 h}{h^2 + K_1 h + K_1 K_2}, \qquad \alpha_2(h) = \frac{K_1 K_2}{h^2 + K_1 h + K_1 K_2}.$$

Thus, $[\mathrm{CO_2(aq)}] = \alpha_0 \mathrm{DIC}$, $[\mathrm{HCO_3^-}] = \alpha_1 \mathrm{DIC}$, $[\mathrm{CO_3^{2-}}] = \alpha_2 \mathrm{DIC}$, and $[\mathrm{OH^-}] = K_w/h$.

Substitution into the alkalinity balance gives

$$\mathrm{DIC}\,[\alpha_1(h) + 2\alpha_2(h)] + \frac{K_w}{h} - h - \mathrm{Alk} = 0.$$

At specified DIC and alkalinity, the positive solution for h determines pH and the dissolved CO_2 concentration,

$$f(\mathrm{DIC}; \mathrm{Alk}) = \alpha_0(h)\,\mathrm{DIC}.$$

Conversely, a prescribed target pH gives $h_{\mathrm{tar}} = c^\circ 10^{-\mathrm{pH_{tar}}}$ and

$$\mathrm{DIC_{tar}} = \frac{\mathrm{Alk} - K_w/h_{\mathrm{tar}} + h_{\mathrm{tar}}}{\alpha_1(h_{\mathrm{tar}}) + 2\alpha_2(h_{\mathrm{tar}})}.$$

Using the inlet-gas reference $C_{\mathrm{in}}^* = HPy_{\mathrm{CO_2,in}}$, the equilibrium state and a target specified by δ_{tar} satisfy

$$f\big(\mathrm{DIC_{eq}}; \mathrm{Alk}\big) = C_{\mathrm{in}}^*, \qquad f(\mathrm{DIC_{tar}}; \mathrm{Alk}) = (1 - \delta_{\mathrm{tar}}) C_{\mathrm{in}}^*.$$

The same closure therefore converts between pH, DIC, and the remaining driving-force fraction. Only targets satisfying $\mathrm{DIC}_0 < \mathrm{DIC}_{\mathrm{tar}} < \mathrm{DIC}_{\mathrm{eq}}$ are considered. Numerical solution settings are given in Appendix B.1.

A.2. Derivation of the Equilibrium-Aware NTU Time and Chemical Burden Factor

At a fixed tank state, the equilibrium-aware model assigns the uniform liquid concentration $c_L = f(\mathrm{DIC}; \mathrm{Alk})$ throughout the contactor. Let $a \in [0, A]$ denote cumulative outer membrane area along the gas-flow direction. With constant volumetric gas flow and the prescribed liquid-basis coefficient K, the gas balance is

$$Q_g \frac{dC_g}{da} = -K\big(mC_g - c_L\big), \qquad C_g(0) = C_{g,\mathrm{in}}, \qquad m = HRT.$$

Solving this linear equation gives

$$C_g(a) = \frac{c_L}{m} + \left(C_{g,\mathrm{in}} - \frac{c_L}{m}\right) e^{-\frac{mKa}{Q_g}}.$$

Since $C^*_{\mathrm{in}} = mC_{g,\mathrm{in}}$ and $\mathrm{NTU}_g = mKA/Q_g$, the instantaneous module absorption rate is

$$\dot{n}_{\mathrm{EA}} = Q_g\big[C_{g,\mathrm{in}} - C_g(A)\big] = Q_g C_{g,\mathrm{in}}\big(1 - e^{-\mathrm{NTU}_g}\big)\left(1 - \frac{c_L}{C^*_{\mathrm{in}}}\right) = \dot{n}_{PS}\,\delta(\mathrm{DIC}).$$

Substituting this rate into the tank balance, $V_L\, d\,\mathrm{DIC}/dt = \dot{n}_{\mathrm{EA}}$, and integrating over the loading interval yields

$$t_{\mathrm{EA}} = \frac{V_L}{\dot{n}_{PS}} \int_{\mathrm{DIC}_0}^{\mathrm{DIC}_{\mathrm{tar}}} \frac{d\,\mathrm{DIC}}{\delta(\mathrm{DIC})}.$$

The corresponding perfect-sink time is $t_{PS} = V_L(\mathrm{DIC}_{\mathrm{tar}} - \mathrm{DIC}_0)/\dot{n}_{PS}$. Factoring out this quantity gives

$$t_{\mathrm{EA}} = t_{PS} B_{\mathrm{chem}}, \qquad B_{\mathrm{chem}} = \frac{1}{\mathrm{DIC}_{\mathrm{tar}} - \mathrm{DIC}_0} \int_{\mathrm{DIC}_0}^{\mathrm{DIC}_{\mathrm{tar}}} \frac{d\,\mathrm{DIC}}{1 - f(\mathrm{DIC}; \mathrm{Alk})/C^*_{\mathrm{in}}}.$$

The multiplicative form is therefore an exact algebraic factorization under the uniform-liquid and constant-parameter assumptions. The factor B_{chem} averages the inverse normalized driving force over the entire DIC loading interval. For the reachable absorption states considered here, $B_{\mathrm{chem}} \geq 1$ and approaches unity when $f(\mathrm{DIC}; \mathrm{Alk}) \ll C^*_{\mathrm{in}}$ throughout the interval.

The absence of liquid-flow dependence follows from the uniform-liquid approximation; finite-pass effects are treated in Section 2.4 and Appendix C.3.

Appendix B. Numerical Implementation and Verification Details

B.1. Solver Settings and Loading-Time Integration

The liquid-pass and gas-coordinate balances in Section 2.4 are integrated by fourth-order Runge–Kutta (RK4) methods. The number of gas-coordinate integration steps, N_{x}, is a numerical resolution parameter for the continuous-row model. At every gas RK4 stage, each liquid pass is integrated from the current tank DIC with the stage's gas composition held fixed. For N_{x} gas steps and liquid-resolution multiplier r_l, the liquid-pass step count is

$$N_l = r_l \max\left[4, \left\lceil \frac{(KA/Q_l) f'(\mathrm{DIC_{eq}}; \mathrm{Alk})}{0.35} \right\rceil\right].$$

Here, f' is evaluated at fixed alkalinity, and $\mathrm{DIC_{eq}}$ is the inlet-gas equilibrium state. This heuristic selects the liquid-pass resolution.

The archived carbonate calculation uses six Newton updates in $\ln(h/c^\circ)$ to solve the full alkalinity balance in Appendix A.1, initialized with the positive root of

$$(\mathrm{Alk} - \mathrm{DIC})h^2 + [(\mathrm{Alk} - 2\mathrm{DIC})K_2 - K_w]h - K_w K_2 = 0.$$

The initializer requires $\mathrm{DIC} < \mathrm{Alk}$, which holds throughout the reported calculations. Equilibrium and δ_{tar} targets use a corresponding six-update solve at prescribed dissolved CO_2. The distributed implementation retains these fixed updates and additionally rejects nonfinite or unconverged chemical states.

For the loading-time integral, define $g_0 = \mathrm{DIC_{eq}} - \mathrm{DIC_0}$ and

$$\eta = \ln\left(\frac{g_0}{\mathrm{DIC_{eq}} - \mathrm{DIC}}\right), \qquad \mathrm{DIC}(\eta) = \mathrm{DIC_{eq}} - g_0 e^{-\eta}.$$

Gauss–Legendre quadrature with N_q nodes is applied to

$$t_i = V_L \int_0^{\eta_{\mathrm{tar}}} \frac{g_0 e^{-\eta}}{\dot{n}_i[\mathrm{DIC}(\eta)]}\, d\eta, \qquad \eta_{\mathrm{tar}} = \ln\left(\frac{g_0}{\mathrm{DIC_{eq}} - \mathrm{DIC_{tar}}}\right), \qquad i \in \{EA, FP\}.$$

The perfect-sink time is evaluated directly. Table B1 specifies the endpoint and trajectory resolutions.

Table B1. Resolution Settings for Endpoint and Trajectory Calculations.

Calculation	Quadrature nodes	Gas steps	Liquid multiplier
Coarse refinement calculation	48	16	1
Fine calculation; Figure 3 and the 100-case set	96	32	2
Further refinement of 10 selected cases	192	64	4
Figure 2 endpoints and pH trajectories	128	32	2
Refinement of 32 selected trajectory states	256	64	4

B.2. Refinement Cases and Trajectory Resolution

Table B2. Complete Refinement-Case Specification and Maximum Coarse-to-Fine Time Differences.

Scenario group	Parameter values	$Q_l/Q_{l,\mathrm{ref}}$	Case–target combinations	Maximum difference (%)
Liquid flow	Baseline chemistry and gas flow	0.03, 0.1, 0.3, 1, 3, 10	30	1.23×10^{-3}
Alkalinity	Alk $= 50$, 200 mol m^{-3}	0.1, 1	20	1.29×10^{-3}
Inlet CO_2	$y_{CO_2,\mathrm{in}} = 200$, 800 ppm	0.1, 1	20	1.31×10^{-3}
Initial loading	$\mathrm{DIC}_0 = 40$, 50 mol m^{-3}	0.1, 1	20	8.12×10^{-4}
Gas flow	$\mathrm{NTU}_g = 0.5, 2$	1	10	6.63×10^{-5}

Five scenarios are further evaluated at $\delta_{\mathrm{tar}} = 0.05$ and 0.01; Table B3 contains all ten comparisons. Overall maxima are summarized in Table 2. These joint refinement checks quantify loading-time sensitivity to the combined resolutions.

Table B3. Absolute Fine-to-Further-Refined Differences in t_{FP}.

Scenario; other inputs at baseline	$\delta_{\mathrm{tar}} = 0.05$ difference (%)	$\delta_{\mathrm{tar}} = 0.01$ difference (%)
$Q_l/Q_{l,\mathrm{ref}} = 0.03$	5.69×10^{-5}	6.71×10^{-5}
Baseline	7.98×10^{-9}	1.36×10^{-7}
Alk $= 50$ mol m^{-3}, $Q_l/Q_{l,\mathrm{ref}} = 0.1$	5.30×10^{-5}	7.15×10^{-5}
Inlet CO_2 = 800 ppm, $Q_l/Q_{l,\mathrm{ref}} = 0.1$	5.35×10^{-5}	7.22×10^{-5}
$\mathrm{NTU}_g = 2$ through a change in Q_g	3.80×10^{-6}	3.66×10^{-6}

The archived Figure 2 trajectories contain 1,200 positive-time states plus the initial state, including the four target pH values. Each time is independently integrated from the initial DIC. Joint refinement at 32 selected states gives a maximum relative time difference of 9.74×10^{-12}. A separate every-other-point sampling check gives a maximum interpolation difference of 5.37×10^{-5} pH units. These assess integration and plotting resolution, respectively.

B.3. Conservation and Benchmark Checks

The chemical closure is compared with independent pH bisection at 101 DIC values from zero to baseline equilibrium. Bisection uses $[0, 16]$, absolute and relative interval-width tolerances of 10^{-12}, and at most 200 iterations. The maximum dissolved-CO_2 difference is 1.15×10^{-13} mol m^{-3}; inverse conversion gives a maximum DIC round-trip difference of 2.13×10^{-14} mol m^{-3}.

For a liquid pass at fixed inlet-gas composition, a 64-step RK4 calculation is compared with the integral balance

$$\int_{\mathrm{DIC_{in}}}^{\mathrm{DIC_{out}}} \frac{d\mathrm{DIC}}{C_{\mathrm{in}}^{*} - f(\mathrm{DIC}; \mathrm{Alk})} = \frac{KA}{Q_l}.$$

The independent solver uses a logarithmic distance-to-equilibrium transformation, adaptive Simpson quadrature with absolute and relative tolerances of 10^{-8} and maximum subdivision depth 24, and outlet-coordinate bisection with absolute and relative tolerances of 10^{-10} and at most 200 iterations. Nine conditions combine $Q_l/Q_{l,\mathrm{ref}} = 0.1$, 1, and 10 with $\mathrm{DIC_{in}}/\mathrm{DIC_{eq}} = 0$, 0.7, and 0.99.

An analytical benchmark replaces carbonate chemistry with $f(\mathrm{DIC}) = \kappa \mathrm{DIC}$, where $\kappa = 0.003$. With

$$\lambda = \frac{KA\kappa}{Q_l}, \qquad \varphi = \frac{1 - e^{-\lambda}}{\lambda},$$

the exact module absorption rate is

$$\dot{n}_{\mathrm{exact}} = Q_g C_{g,\mathrm{in}} \left(1 - \frac{\kappa \mathrm{DIC}}{C_{\mathrm{in}}^{*}}\right)\left[1 - e^{-\mathrm{NTU}_g \varphi}\right].$$

The comparison uses baseline transport conditions, tank DIC values of 0, 1, and 3 mol m^{-3}, $N_{\mathrm{x}} = 32$, and $r_l = 4$.

Finally, assigning tank composition uniformly throughout the liquid side recovers t_{EA}. This check combines $Q_l/Q_{l,\mathrm{ref}} = 0.1$, 1, and 10 with $\delta_{\mathrm{tar}} = 0.15$, 0.05, and 0.01, using $(N_q, N_{\mathrm{x}}, r_l) = (128,64,4)$ without gas-flow contraction. This limiting-model consistency check uses the same chemistry and time quadrature.

Appendix C. Supplementary Comparisons and First-Order Pass Correction

C.1. Additional Chemical Conditions

Table C1 compares additional chemical conditions at $\delta_{\mathrm{tar}} = 0.05$ and two liquid flow rates, using $E_{\mathrm{EA}} = 100(t_{FP}/t_{\mathrm{EA}} - 1)$. The complete scenario set is defined in Table B2; geometry and the prescribed K remain fixed. A common δ_{tar} represents the same normalized remaining driving force, while target pH, loading capacity, and loading time vary with chemistry.

Table C1. Additional Loading Time Relative to the Equilibrium-Aware Prediction at $\delta_{\mathrm{tar}} = 0.05$.

Alkalinity (mol m⁻³)	Inlet CO_2 (ppm)	Initial DIC (mol m⁻³)	E_{EA} (%) at $Q_l/Q_{l,\mathrm{ref}} = 0.1$	E_{EA} (%) at $Q_l/Q_{l,\mathrm{ref}} = 1$
100	400	0	13.38	1.13
50	400	0	29.38	2.15
200	400	0	6.75	0.61
100	200	0	6.75	0.61
100	800	0	29.43	2.16
100	400	40	22.21	1.88
100	400	50	26.54	2.25

Within these cases, lower alkalinity, higher inlet CO_2, and lower liquid flow increase the relative finite-pass contribution. Chemical conditions affect both the CO_2–DIC relation and the loading interval.

For initial loading, the relative and absolute effects differ. At baseline liquid flow, increasing DIC_0 from 0 to 50 mol m⁻³ reduces t_{EA} from 4.145 to 2.085 h, while $t_{FP} - t_{\mathrm{EA}}$ remains approximately 0.0469 h. The increase in E_{EA} from 1.13% to 2.25% therefore mainly reflects the shorter remaining loading time.

C.2. Effect of Gas-Flow Contraction

A paired calculation replaced the constant gas volumetric flow with the small composition-dependent contraction implied by CO_2 removal at fixed temperature and pressure. Across nine baseline-chemistry comparisons, the maximum absolute loading-time change was 1.05×10^{-3}%. This supports the constant-flow approximation for the reported 400 ppm cases. The paired values are retained in the companion numerical records.

C.3. First-Order Correction for Liquid-Pass Loading

A compact correction can be obtained by expanding the liquid-pass balance about the current tank DIC. The derivation retains constant gas volumetric flow and the gas-sequential/parallel-liquid arrangement. At a fixed tank state, define $\beta = KA/Q_l$ and $G(x) = mC_g(x) - f(\mathrm{DIC};\mathrm{Alk})$. Expansion of the pass outlet gives

$$\mathrm{DIC}_{\mathrm{out}}(x) - \mathrm{DIC} \approx \beta G(x)[1 - \frac{\beta}{2} f'(\mathrm{DIC};\mathrm{Alk})].$$

Here, f' is the derivative at fixed alkalinity. The expansion requires a small change in driving force within each pass and an adequate local linear approximation to the CO_2–DIC relation. In particular, $(KA/Q_l)f'(\mathrm{DIC};\mathrm{Alk})$ should remain small over the loading interval.

Substituting the pass expansion into the gas balance and retaining the first correction in $1/Q_l$ yields

$$\dot{n}_{FP}(\mathrm{DIC}) \approx \dot{n}_{\mathrm{EA}}(\mathrm{DIC}) \left[1 - \frac{\mathrm{NTU}_g e^{-\mathrm{NTU}_g}}{1 - e^{-\mathrm{NTU}_g}} \frac{KA}{2Q_l} f'(\mathrm{DIC};\mathrm{Alk})\right].$$

Expanding the inverse rate to the same order and integrating the tank balance gives the additional loading time

$$\Delta t_1 = \frac{V_L}{2Q_l} \frac{\mathrm{NTU}_g^2 e^{-\mathrm{NTU}_g}}{\left(1 - e^{-\mathrm{NTU}_g}\right)^2} \int_{\mathrm{DIC}_0}^{\mathrm{DIC}_{\mathrm{tar}}} \frac{f'(\mathrm{DIC};\mathrm{Alk})}{C_{\mathrm{in}}^* - f(\mathrm{DIC};\mathrm{Alk})} d\mathrm{DIC}.$$

Since $\delta(\mathrm{DIC}) = 1 - f(\mathrm{DIC};\mathrm{Alk})/C_{\mathrm{in}}^*$, the integral equals $\ln(\delta_0/\delta_{\mathrm{tar}})$, where $\delta_0 = \delta(\mathrm{DIC}_0)$. The corrected estimate is therefore

$$t_{\mathrm{corr}} = t_{\mathrm{EA}} + \Delta t_1,$$

$$\Delta t_1 = \frac{V_L}{2Q_l} \frac{\mathrm{NTU}_g^2 e^{-\mathrm{NTU}_g}}{\left(1 - e^{-\mathrm{NTU}_g}\right)^2} \ln\left(\frac{\delta_0}{\delta_{\mathrm{tar}}}\right).$$

This additive loading-time correction follows from a first-order Taylor expansion of the coupled balances.

For the correction comparison, define

$$\varepsilon_{\mathrm{corr}} = 100\left(\frac{t_{\mathrm{corr}}}{t_{FP}} - 1\right).$$

Unlike E_{EA} in the main text, this metric uses t_{FP} as the denominator; negative values indicate a remaining underprediction. Table C2 reports the baseline-chemistry flow sweep at $\delta_{\mathrm{tar}} = 0.05$, for which $t_{\mathrm{EA}} = 4.144937$ h at every flow rate.

Table C2. First-Order Corrected Loading Times at Baseline Chemistry and $\delta_{\mathrm{tar}} = 0.05$.

$Q_l/Q_{l,\mathrm{ref}}$	t_{FP} (h)	t_{corr} (h)	$\varepsilon_{\mathrm{corr}}$ (%)
0.03	6.742414	5.677436	−15.7952
0.1	4.699604	4.604687	−2.01968
0.3	4.308461	4.298187	−0.238461
1	4.191826	4.190912	−0.0218005
3	4.160363	4.160262	−0.00243181
10	4.149544	4.149535	−0.000219150

The correction reduces the absolute time discrepancy relative to t_{EA} in all 100 case–target combinations in Table B2. Nevertheless, appreciable errors remain at low liquid flow: the underprediction is 15.80% at $Q_l/Q_{l,\mathrm{ref}} = 0.03$ and $\delta_{\mathrm{tar}} = 0.05$, reaching 16.82% at $\delta_{\mathrm{tar}} = 0.01$.

Separate high-flow limiting checks at baseline chemistry and $\delta_{\mathrm{tar}} = 0.05$ give $(t_{FP} - t_{\mathrm{EA}})/\Delta t_1 = 1.01988$ at $Q_l/Q_{l,\mathrm{ref}} = 1$ and 1.00002 at $Q_l/Q_{l,\mathrm{ref}} = 1000$, consistent with recovery of the leading correction as within-pass loading diminishes. Both limiting checks retain fixed K.

Appendix D. Code, Input Records, and Reproducibility

The original calculations used Python 3.12.14, NumPy 2.5.2, and Matplotlib 3.11.1 on Windows. The package's config.toml records the full-precision baseline, study sets, and numerical settings; the baseline liquid flow is $4.523893421169301 \times 10^{-6}\ m^3\ s^{-1}$. Historical inputs and source identities are retained in results/reference/provenance.json, with file hashes in the reference manifest. The common implementation is in hfmc/, and dependency versions are pinned in requirements.txt.

The package stores the full-precision configuration, saved numerical results, verification records, and source hashes under results/reference. Its README describes installation, execution, and checkpoints. The notebooks display stored results by default; explicit recomputation writes separate outputs.